



Supervised Contrastive Learning for Vietnamese Fine-Grained Emotion Classification on Social Media Text

Nghiên cứu học tương phản có giám sát trong phân loại cung bậc cảm xúc tiếng Việt trên mạng xã hội

20,664

comments · 27 emotions + neutral
(ViGoEmotions benchmark)



ViSoBERT + JSCL

Vietnamese social encoder +
Jaccard contrastive loss



+1.93 / +1.05

Macro-P / Macro-F1 gain of
JSCL vs. baseline (Recall held)



WHY?

The challenge

- ◆ **Noise & brevity** — comments are short and messy.
- ◆ **Multi-label overlap** — ~6.9 labels per sample on average.
- ◆ **High inter-class similarity** — near-synonym emotions (sad vs. despair) blur together.

Plain CE/BCE losses optimize only the decision boundary; they never shape the **geometry** of the representation space. No prior work unites a Vietnamese social encoder, a multi-label objective, and a highly overlapping label set — the gap we target.

WHAT?

The task

Multi-label, fine-grained emotion classification of Vietnamese social-media text.

- ◆ **Input:** a raw comment, possibly mixing slang, teencode, emoji.
- ◆ **Output:** a multi-hot vector over **28 labels** — 27 emotions + neutral.
- ◆ A comment may carry several emotions at once.

OBJECTIVES

Three goals

- ◆ **Compare variants.** Strict SCL vs. Jaccard JSCL on ViGoEmotions across six metrics (Subset Acc, Macro P/R/F1, Weighted F1, Micro F1).
- ◆ **Build the framework.** Co-train a BCE classifier with supervised contrastive learning on a ViSoBERT backbone.
- ◆ **Analyze geometry & errors.** t-SNE of embeddings + confusion on confusable pairs (anger↔annoyance, joy↔amusement).

INSIGHT

Overlap-aware pairing is the lever

The decisive factor is **how positive pairs are defined**. Treating any shared-label pair as fully positive over-separates the space and hurts macro performance. A **soft, overlap-proportional pull (Jaccard)** yields cohesive, interpretable clusters — independent emotions like *gratitude* stay compact while related ones stay near — at almost no extra cost.

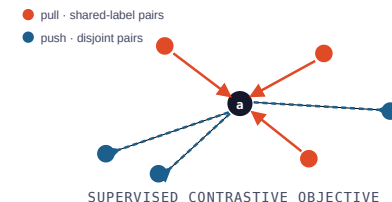
KEY STATS

OVERVIEW

A geometric regularizer for emotion space

Vietnamese social platforms generate enormous volumes of comments mirroring public sentiment in real time. Detecting **fine-grained emotions** — not just positive vs. negative — is key to automated customer care, brand-reputation management, and opinion analysis.

We pair **ViSoBERT** (pre-trained on Vietnamese social text) with a **supervised contrastive loss** that regularizes the representation space, comparing two ways to define positive pairs: strict SCL vs. Jaccard-weighted JSCL.



METHODOLOGY

Joint objective on a two-view encoder

$$\mathcal{L}_{\text{Joint}} = \alpha \cdot \mathcal{L}_{\text{BCE}} + (1 - \alpha) \cdot \mathcal{L}_{\text{SupCon}}$$

- ◆ **Backbone — ViSoBERT** (~100M): robust to teencode/emoji, fits a single T4 GPU.
- ◆ **Two-view:** each comment passes the encoder twice under dropout → two semantically identical views (no manual augmentation). [CLS] (768-D) splits into a classifier head (28 logits → BCE) and a projection head (128-D, L2 → contrastive).
- ◆ **SCL (strict):** any pair sharing ≥ 1 label is fully positive.
- ◆ **JSCL:** soft positive weights by Jaccard label overlap.

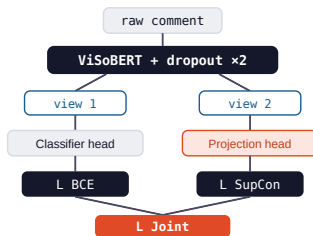


FIG. 1 — END-TO-END PIPELINE

Data: ViGoEmotions
Optimizer: AdamW
Batch: 16
Hardware: Colab T4

Size: 20,664
LR: 2×10^{-5}
Epochs: 5
Proj: 128 (L2)

RESULTS

Jaccard wins where strict separation fails

Headline: JSCL is the only variant that raises Precision *without* sacrificing Recall — improving Macro Precision, Subset Accuracy, and Macro F1 over the BCE baseline while keeping Recall steady. Strict SCL over-separates: Precision up, Recall down, F1 falls.

VALIDATION SET

Model	Subset	M-P	M-R	M-F1	W-F1	μ -F1
BCE base	.2633	.6595	.4521	.5271	.5523	.5653
+ SCL	.2546	.6835	.4210	.5056	.5382	.5571
+ JSCL (BEST)	.2643	.6788	.4574	.5376	.5618	.5737

TEST SET

Model	Subset	M-P	M-R	M-F1	W-F1	μ -F1
BCE base	.2535	.6634	.4629	.5382	.5647	.5762
+ SCL	.2477	.6650	.4308	.5116	.5450	.5624
+ JSCL (BEST)	.2550	.6712	.4551	.5345	.5610	.5731

Baseline · one mixed blob



FIG. 2 — T-SNE OF THE EMOTION SPACE

Error hotspots — twin pairs: anger↔annoyance (59), disgust↔annoyance (58), joy↔amusement (31), sadness↔disappointment (25); sarcasm + emoji remain hard.

CONCLUSION

Takeaway & next steps

A **label-overlap-aware** definition of positive pairs is an effective, resource-efficient route to multi-label fine-grained emotion recognition for Vietnamese.

- ◆ Tighter intra-class cluster constraints (CES-CL).
- ◆ Model sarcasm & emoji with auxiliary signals.
- ◆ Larger batch/epoch budgets & multi-seed runs for robustness.

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